Web and HTTP

Web and HTTP: Quick Review

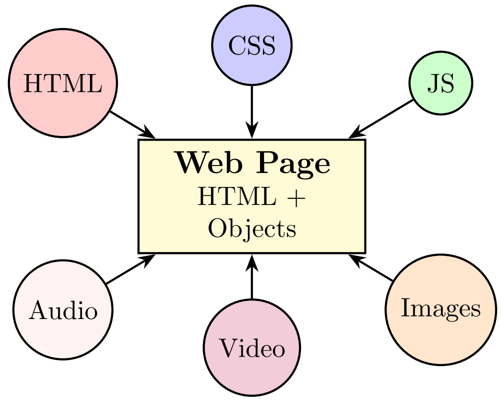
Key Concepts of Web Pages

- A **web page** consists of multiple **objects**, often stored on different web servers.
- An **object** can be: HTML file, CSS file, JavaScript file, image, audio, video, etc.
- A complete web page = **base HTML file** + referenced objects.
- Each object is uniquely identified by a **URL (Uniform Resource Locator)**.
- Example: `https://www.someschool.edu:8080/someDept/pic.gif`
  - **Protocol:** https
  - **Host name:** www.someschool.edu
  - **Port:** 8080 (optional, defaults: 80 for HTTP, 443 for HTTPS)
  - **Path:** /someDept/pic.gif

What is the World Wide Web?

The **World Wide Web (WWW)** is a distributed system of information exchange consisting of:

- **Web pages** containing text, images, multimedia, and interactive elements
- **Web servers** that store and serve these pages
- **Web browsers** that request and render content for users
- **Hyperlinks** that interconnect documents globally



HTTP Overview

Definition

HTTP (Hypertext Transfer Protocol) is the Web's application-layer protocol. It specifies:

- How clients and servers **format and exchange messages**
- What **actions** are taken for each request/response
- The **syntax and semantics** of communication

HTTP Architecture

HTTP is based on the **client-server model**:

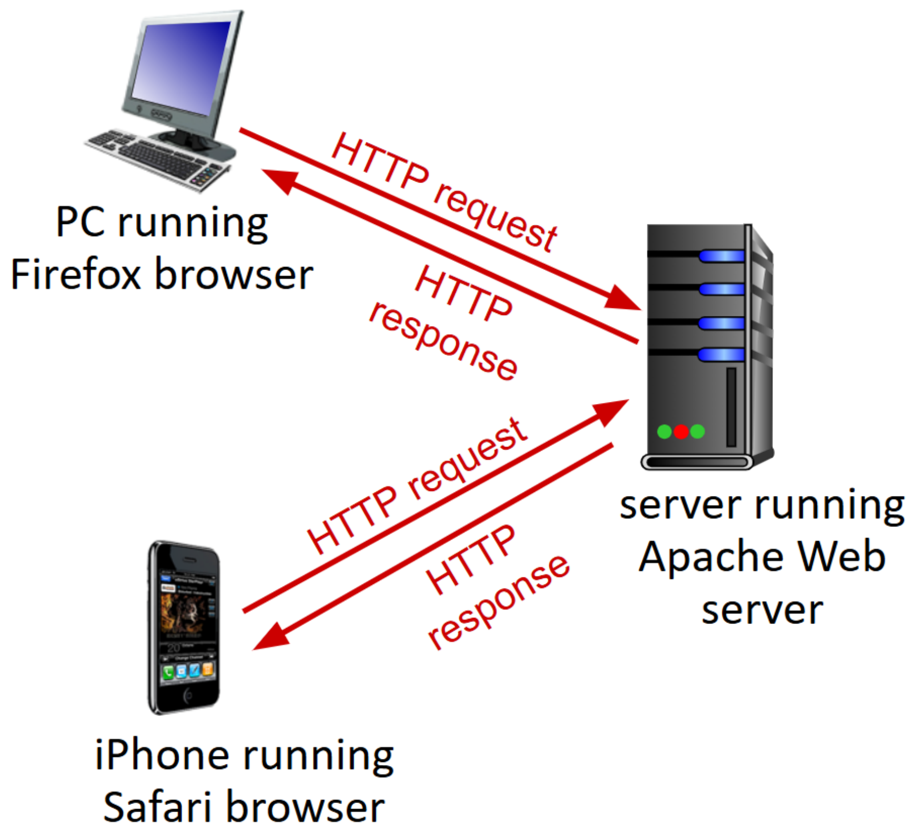


Figure 1: HTTP Client-Server Architecture

HTTP and TCP

HTTP runs on top of TCP to provide reliable data transfer:

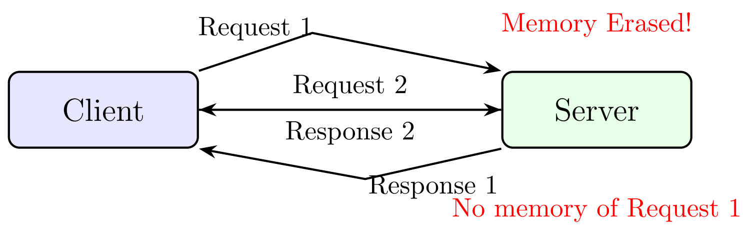
Application Layer (HTTP)	Request/Response Messages
Transport Layer (TCP)	Reliable Data Transfer
Network Layer (IP)	Packet Routing

Why TCP for HTTP?

- **Reliability:** Ensures error-free delivery
- **Ordering:** Guarantees correct sequence of messages
- **Flow Control:** Adjusts data flow between sender/receiver
- **Congestion Control:** Prevents overload in the network

HTTP Statelessness

HTTP is stateless: The server keeps no memory of past requests.



Implications of Statelessness

- **Simplicity:** Server design is lighter
- **Reliability:** Server crashes do not lose client data
- **Scalability:** Supports many users efficiently
- **Challenge:** Applications needing state use *cookies, sessions, or tokens*

Conclusion

- The WWW is a vast distributed system built on web pages, objects, and hyperlinks.
- HTTP is the communication backbone, running on TCP to ensure reliable data transfer.
- Statelessness keeps HTTP simple and scalable, but applications use cookies/sessions when state is needed.